INFLUENCE OF WEATHER PATTERNS ON STOCK MARKET PRICE

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1. Research Methodology

1.1 Introduction

I have always had a passion for mathematics and its use in a real-world scenario, particularly in regard to the role outside forces may have on financial markets. I remember being a student interested in environmental issues and the financial markets and wondering about the existence of a scientific approach that could relate to natural factors with some of the fundamental drivers of stock prices. This concern has heightened of late in relation to the increasing effects of climatic change to various sectors of the economy with a good example being the energy sector which relies heavily on the ecological environment.

1.2 Aim

My research will attempt to demonstrate whether in fact, the change in weather patterns and average temperature has a significant effect on the market fluctuations of companies within the energy sector. This would entail the use of weather information from NOAA and stock prices sourced from Yahoo finance covering a period of at least five years. Equipped with an assortment of statistical tools through correlation and regression analysis together with hypothesis testing, I will be able to explore the relationship that exists between these external factors and stock price volatility and whether such relationships can be relied on in making financial predictions.

1.3 Rationale

This topic appeared to be interesting because of practical applicability and richness of data that is relevant. I hope that by studying this intersection between mathematics, environmental science and finance, I might be able to uncover some interesting

connections between weather and financial markets. This research integrates some of my academic interests and responses to two very pressing challenges: climate change and economic growth. With global warming threats and the focus on climate change risk factors, weather risk modeling and practical applications in financial engineering and risk management are very promising.

2. Data and Methods

2.1 Data Sources

The primary data used in this research was gathered from two main sources: weather data from NOAA and stock prices from Yahoo Finance. The weather data is comprised of the average temperature every day over a span of 2 years, from January 1st, 2020 to December 31st, 2021, for three regions: New Jersey, Wilmington, and Philadelphia. Closing stock prices were obtained for three companies associated with these regions:

PEG (Public Service Enterprise Group) for New Jersey

EXC (Exelon Corporation) for Wilmington

SUN (Sunoco) for Philadelphia

This data was collected over a two-year period to provide sufficient coverage for both long-term trends and short-term volatility in both temperature and stock prices.

2.2 Statistical Techniques

The analysis includes several important statistical methods, each of which is described below. Among them are correlation analysis, quantile regression, and the

calculation of elasticity. In this section, I explain each method theoretically and apply it to the gathered data.

2.2.1 Correlation Analysis

Correlation quantifies the degree of linear relationship between two variables, that is, the extent to which an alteration in one variable brings about changes in another. In this context, it is important to see the correlation between temperature and stock price since this gives first hints of whether weather has an effect on business activities. A high correlation might mean that it can be predicted with respect to time, whereas a low correlation could imply that stock prices may be affected by prevailing market sentiments or other economic changes. It is quantified using the Pearson correlation coefficient, denoted by r, which is given by the formula:

$$r = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2 \sum((Y_i - \bar{Y})^2}}$$

Within this equation, X_i represents the temperature values at each time points, Y_i represents the corresponding stock prices, and $\bar{X}$ and $\bar{Y}$ are the means of temperature and stock price, respectively. The correlation coefficient r ranges from -1 (perfect negative correlation) to 1 (perfect positive correlation). If r=0, it suggests no linear relationship. For this study, I computed the correlation between the average temperature in each region and the corresponding stock prices of PEG, EXC, and SUN over the two-year period.

2.2.2 Quantile Regression Analysis

Quantile regression provides a fuller analysis than ordinary least squares (OLS) regression because it estimates the relationship between the independent variable of temperature with regard to the dependent variable of stock prices at various percentiles within the stock price distribution. This will enable the analysis to allow for the temperature effect to be identified at more tiers of the dependent variable distribution rather than just at the average (mean). The basic model for quantile regression is given by the formula:

$$Q_Y(\tau|X) = \beta_0(\tau) + \beta_1(\tau)X$$

Within this equation, $Q_Y(\tau|X)$ is the quantile τ of the stock price conditional on the temperature X, $\beta_0(\tau)$ is the intercept for the quantile τ, and $\beta_1(\tau)$ is the slope, showing the change in stock price for a 1-degree change in temperature at quantile τ.

Unlike OLS, which provides a single coefficient for the relationship between variables, quantile regression provides different coefficients for different quantiles. I chose to examine the 25th, 50th (median), and 75th quantiles, which correspond to low, median, and high stock prices, respectively.

2.2.3 ARIMA model for Time Series Forecasting

One of the methods that is most widely used in time series analysis is the ARIMA model which means the Autoregressive Integrated Moving Average. It is a combination of three processes, autoregression, differencing, and moving averages and thus can model trends, seasonal, and cycle variations, as well as erratic influence in the history. Regarding stock price forecasting, the ARIMA model can also be utilized in trend identification with

variables such as temperature serving as external influence factors. With the inclusion of weather factors, it can add valuable information to the stability and volatility of stocks in the energy sector, allowing one to evaluate whether temperature is a significant factor in the situation. Therefore, the ARIMA model is generally identified as the most effective methodology for the forecasting of time series data, and more so in instances where patterns of autocorrelation do exist. Construction of the ARIMA model depends on three specific parameters: p, d, and q, representing the autoregressive, differencing, and moving average components, respectively. The overall structure of the ARIMA model can be expressed as:

$$y_t = c + \emptyset_1 y_{t-1} + \emptyset_2 y_{t-2} + \cdots + \emptyset_p y_{t-p} + \theta_{1\varepsilon_{t-1}} + \theta_{2\varepsilon_{t-2}} + \cdots + \theta_{q\varepsilon_{t-q}} + \varepsilon_t$$

Within this equation, y_t is the value of the stock price at time t, $\emptyset_p$ are the autoregressive coefficients, θ_q are the moving average coefficients, ε_t is the white noise error term, and c is the constant term.

The ARIMA model will be fitted on the two-year stock prices datasets for PEG, EXC, and SUN. This will include the differencing aspect, d, for stationarity, if necessary. The best fitting model will be chosen based on Akaike Information Criterion (AIC). Residual diagnostics will be performed to ensure that the errors are uncorrelated, and then the model will be used for out-of-sample predictions in order to determine how weather patterns can influence future stock prices.

2.2.4 Partial Derivatives

Other than the ARIMA methodology, a sensitivity analysis using partial derivatives will be conducted to analyze the effect of small changes in temperature on the stock price. The use of partial derivatives allows an investigation into how only one variable, in this case, temperature, affects the stock price while keeping all other variables constant. This relationship can be described by the following equation showing the partial derivative of stock price S with respect to temperature T:

$$\frac{\partial S}{\partial T} = \log_{\Delta T \to 0} \frac{\Delta S}{\Delta T}$$

Within this equation, $\frac{\partial S}{\partial T}$ represents the rate of change in stock price with respect to temperature, and ΔS and ΔT are small changes in stock price and temperature, respectively.

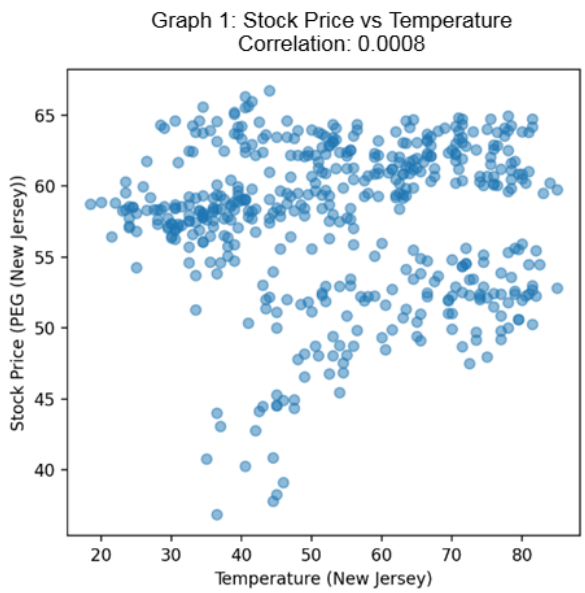
From this derivative, this study will measure the sensitivity of PEG, EXC, and SUN stock prices to changes in the average daily temperature. The result provides a clear meaning about the elasticity of the stock prices relative to changes in temperature, which will allow for more accurate predictions as to how companies operating in climate-sensitive industries may react to changeable weather conditions. This will be especially useful in the identification of whether the relationship between temperature and stock prices is linear, or if larger changes in temperature produce more-than-proportionately larger movements in stock prices.

3. Quantitative Analysis

3.1 Temperature Data and Stock Prices

I applied correlation analysis and quantile regression to investigate the relationship between average temperature and stock prices. In this regard, the first step was to calculate the correlation coefficients that reveal the correlation between temperature and stock prices of PEG, headquartered in New Jersey; EXC, located in Wilmington; and SUN, headquartered in Philadelphia.

3.1.1 Correlation Analysis: PEG (New Jersey)



The correlation coefficient r between PEG's stock price and temperature is 0.0008, given that there are 493 data points for each location and stock, calculated using:

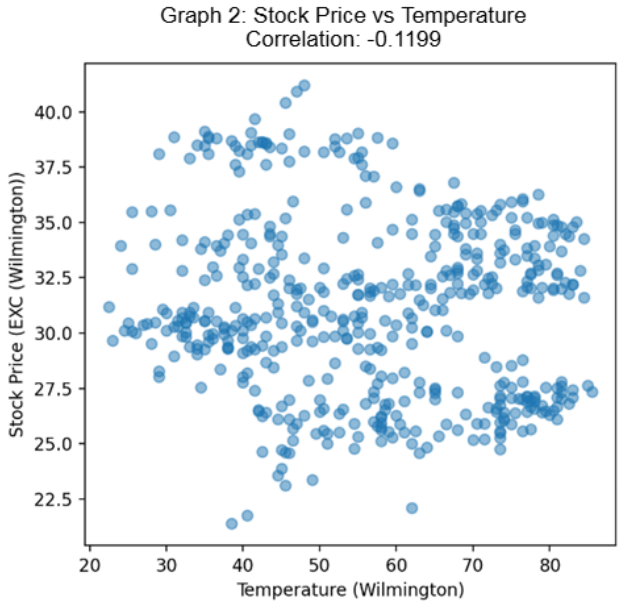
$$r_{PEG} = \frac{\sum_{i=1}^{493}(PEG_i - \overline{PEG})(Temp_{NJ_i} - \overline{Temp_{NJ}})}{\sqrt{\sum_{i=1}^{493}(PEG_i - \overline{PEG})^2}\sqrt{\sum_{i=1}^{493}(Temp_{NJ_i} - \overline{Temp_{NJ}})^2}} = 0.0008$$

This value demonstrates a very weak positive correlation between temperature and PEG's stock price. This finding implies that in regard to changes in the local temperature,

PEG's stock prices are less sensitive, probably because there are many stock performance determinants of a big utility. Short-term contracting or a developed portfolio of energy sources may protect PEG's activities from seasonal factors. Such measures would improve understanding of possible indirect measures involving energy consumption and consumption changes caused by temperature variation. The low correlation also poses the question of how reliance on renewable energy sources affects the PEG financial outcomes in relation to the environmental factors.

PEG primarily operates in energy distribution and builds facilities; thus, the stock prices are often connected with regulatory policies, prices on energy markets, and other economics. This context conforms with the observed temperature insensitivity because the firm may be concerned about its temperature and focus on short term fluctuations of the operations.

3.1.2 Correlation Analysis: EXC (Wilmington)



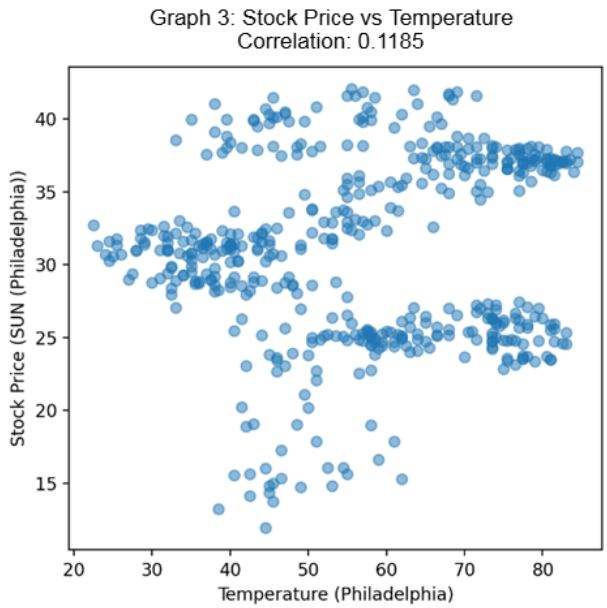
The correlation coefficient r between EXC's stock price and temperature in Wilmington is:

$$r_{PEG} = -0.1199$$

This inverse relation may show growth in EXC's susceptibility as temperature rises. Extreme loads and/or decreased efficiency in certain methods of energy production could lead to increased operational costs. The relationship could also stem from changes arising out of the higher temperatures wherein electricity distribution gets interrupted on extreme weather events, or emissions regulation costs increase; but this relationship is already weak. The weak nature of the relationship suggests that other factors, such as fuel mix, regulatory environment, or broader economic trends, likely play a dominant role. The correlation in greater detail could explain the fluctuations in business by considering seasonal variation in demand in relation to temperature increase.

EXC operates across a region characterized by strong seasonal variability in energy requirements. Summertime extremes could place a strain on Exelon's operations since cooling demand will be at its highest, but there is little prospect of station output because of inadequate tactics, which could lower stock values. Understanding this dynamic could contextualize the observed the stock return trends within the context of the weather variables that appear to drive the changes.

3.1.3 Correlation Analysis: SUN (Philadelphia)



The correlation coefficient r between SUN's stock price and temperature in Philadelphia is:

$$r_{PEG} = 0.1185$$

The observed relationship could indicate a seasonal increase in the use for certain energy commodities like cooling systems and transportation amongst others during the summer months. This may bolster SUN's total income, assuming the firm is well positioned in such markets. But the rather modest strength of the correlation suggests that while temperature is a relevant element it is not the only one. Other aspects like variation in fuel prices or change in the resilience of infrastructure due to weather could complement this understanding.

As for now, SUN's business strategy is focused on fuel distribution especially in suburbs of Philadelphia. This may explain the firm's sensitivity to temperature as it bases

its income on sales of energy products on consumer and industrial sales. Increased temperatures may result in more travel and higher use of cooling systems both of which result in more energy being consumed. Such an operational strategy may help explain the weak positive correction that was apparent in this analysis.

3.2 Regression Models and Equations

3.2.1 Quantile Regression

Quantile regression is applied to analyze the effect of temperature on the varying segments of the distribution. From here, the 25th, 50th-that is, the median-and 75th quantile regression equations are derived.

Company	Quantile 0.25	Quantile 0.50	Quantile 0.75
PEG	-0.0957	0.0400	0.0623
EXC	-0.0606	0.0303	-0.0079
SUN	-0.0873	0.1197	0.1149

The quantitative analysis carried out through quantile regression shows that stock price of the three companies examined will also react differently to temperature changes. PEG (New Jersey) appears to be insensitive overall of its stock price, with a slight negative response at lower quantiles and small positive correlations at higher quantiles. This means that stock price of PEG is affected more by average economic phenomenon rather than the weather in the particular region.

EXC (Wilmington) though, appears to be moderately sensitive at lower quantiles as stock prices are slightly affected as temperature rises and the median and upper quantiles appear to be almost unaffected. This pattern could suggest that the stock performance of EXC is relatively strong and stable due to an array of energy businesses performed by the company and low practices to temperature and other changes in the environment within a short term.

SUN (Philadelphia), on the other hand, appears to react more sensitively towards temperature changes. The stock price at the median and higher quantiles shows positive direction which is indicative of a warm temperature bias. This aligns with the seasonal pattern for earnings of energy goods in warmer periods that features increased returns.

The results stress how the weather and stock price shifts are related when focusing on the energy markets. Since PEG is not sensitive to temperature fluctuations, its stock price is more likely determined by factors such as the state of the energy market or changes in legislation rather than the weather in the short run. This suggests stability of sorts, but may also imply lesser chances of income generation from capitalizing on the climate induced developments.

By contrast, brief shot sensitivity of EXC to temperature makes it clear why dominance of variety is good for the business. Because of preserving stability across quantiles, EXC seems to be less affected by the processes of local weather changes making it ideal for risk averse investors who want to earn stable returns.

SUN is differentiated with a strong expansion coefficient as long as temperature rises which informs the extent season energy utilization affects it. This dependence is both potential and a problem. For instance, warmer climate increases the appeal for its energy products and hence profits. However, it also creates risks for SUN as it makes it susceptible to the changes in climatic conditions.

3.3 ARIMA Model

The ARIMA model selection for each stock was based on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), which balance model complexity with how well it fits the data. This is shown for all three stocks below:

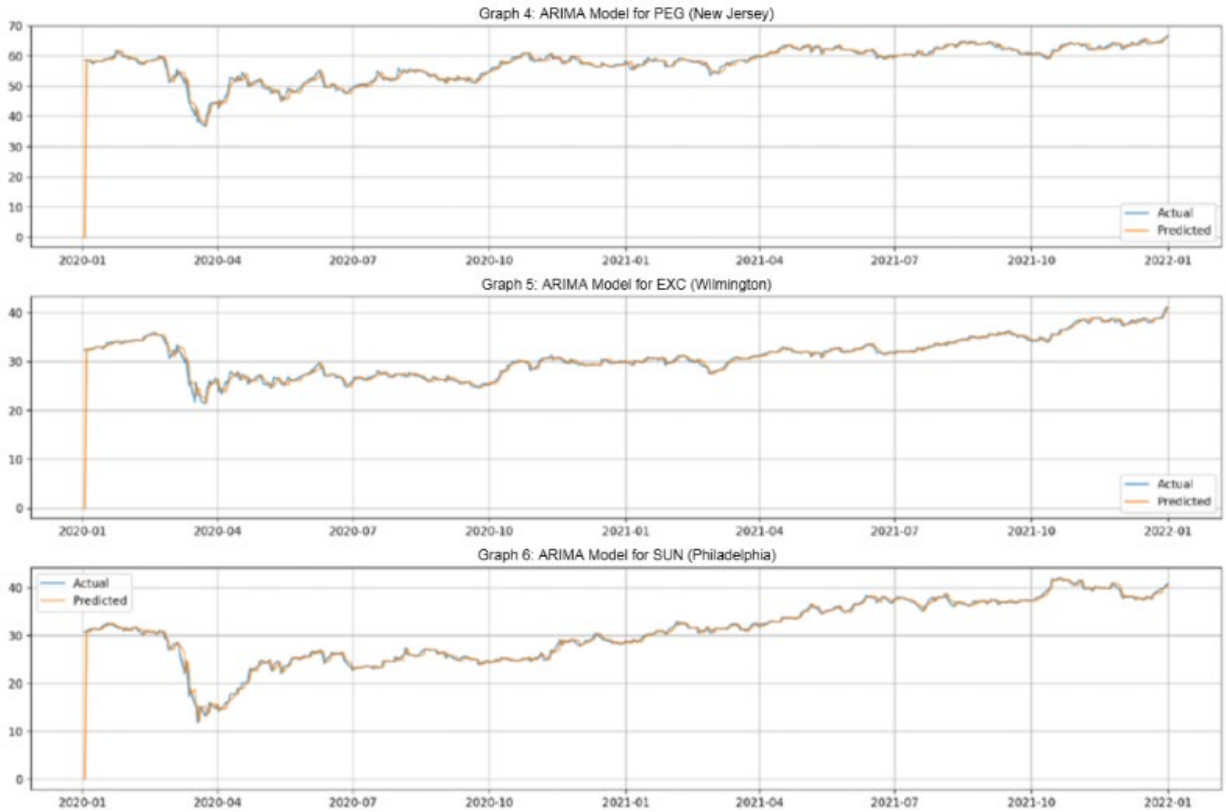
PEG (New Jersey): AIC = 1430.93, BIC = 1443.6

EXC (Wilmington): AIC = 930.4, BIC = 943.07

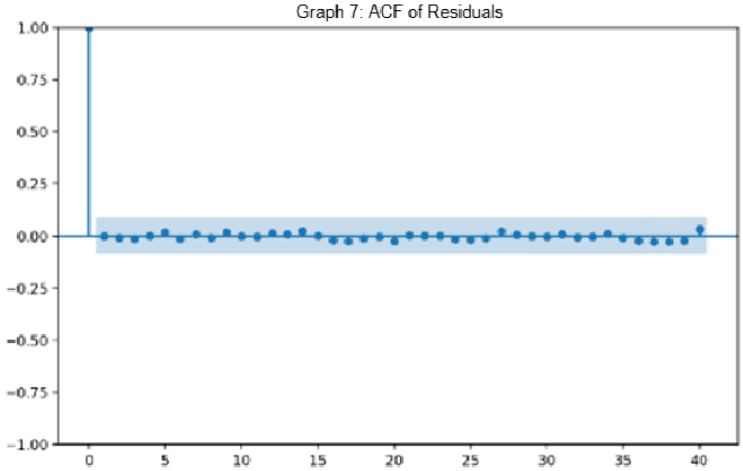
SUN (Philadelphia): AIC = 1060.75, BIC = 1073.41

The reduced AIC and BIC for EXC indicate a better model fit for that stock compared to PEG and SUN. The choice of ARIMA models reflects an autoregressive (AR) and moving average (MA) process for the price of each stock, allowing it to capture both trends and cyclic variations.

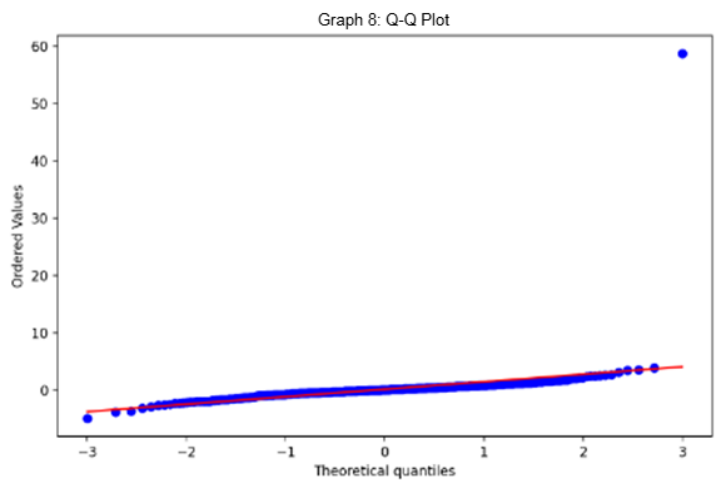
Below is the plot of ARIMA model fits for PEG, EXC, and SUN. The models are useful in visualizing predicted stock prices, shown by an orange line, against actual stock prices represented by a blue line as time goes on. Each of the ARIMA models was uniquely made to model both short-run and long-run patterns in stock price movements.



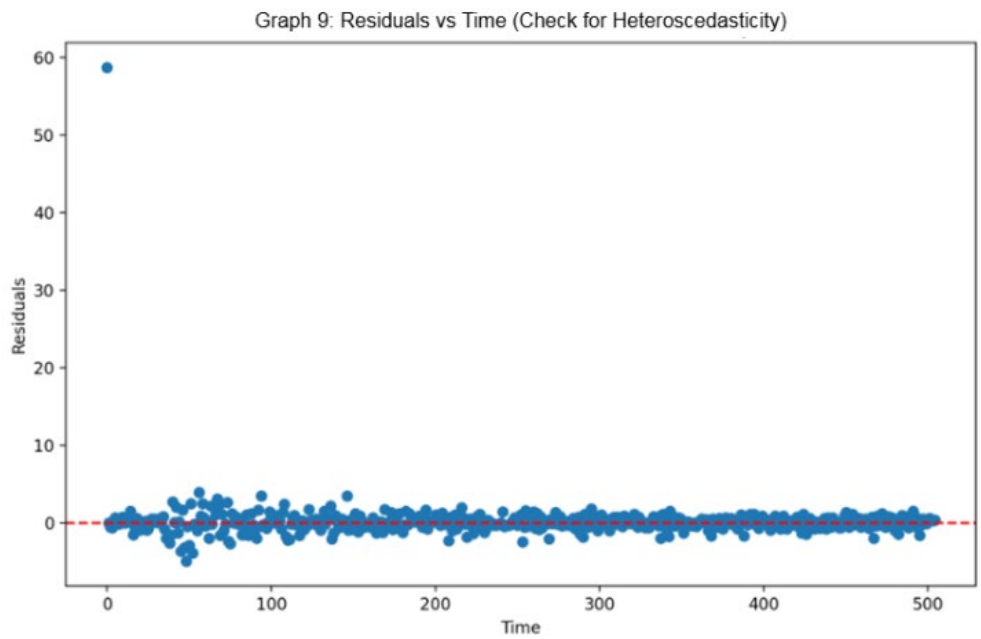
PEG (New Jersey) shows tight alignment between predicted and actual stock prices, except for minor deviations around mid-2020. EXC (Wilmington) displays a similar tight fit, with the model capturing small price fluctuations accurately. SUN (Philadelphia) follows a similar pattern, with the ARIMA model capturing both dips and rises effectively.



One important assumption in time series forecasting models, including ARIMA, is that residuals do not hold any sort of significant correlation. The high p-values close to 1.0, when confirmed through the Ljung-Box test for PEG, EXC, and SUN over different lags of 10, 20, and 30, confirmed uncorrelated residuals. This is indeed confirmed by the Autocorrelation Function (ACF) plot, where most of its correlations fall within the confidence bands, reinforcing the fact that ARIMA models captured the underlying time series patterns.



Although ARIMA models represented time series data effectively, the residuals showed a lack of normal distribution in the Q-Q plot, as depicted by an apparent deviation from the diagonal line. This is further supported by the Jarque-Bera test; with an infinitesimal p-value of 0.0000, the null hypothesis on normality is rejected. Moreover, high kurtosis of 382.57 and skewness of 18.32 reveal that residuals exhibit heavy tails and a right-skewed distribution. Such non-normality will make the actual prediction intervals unreliable for very extreme stock price movements.



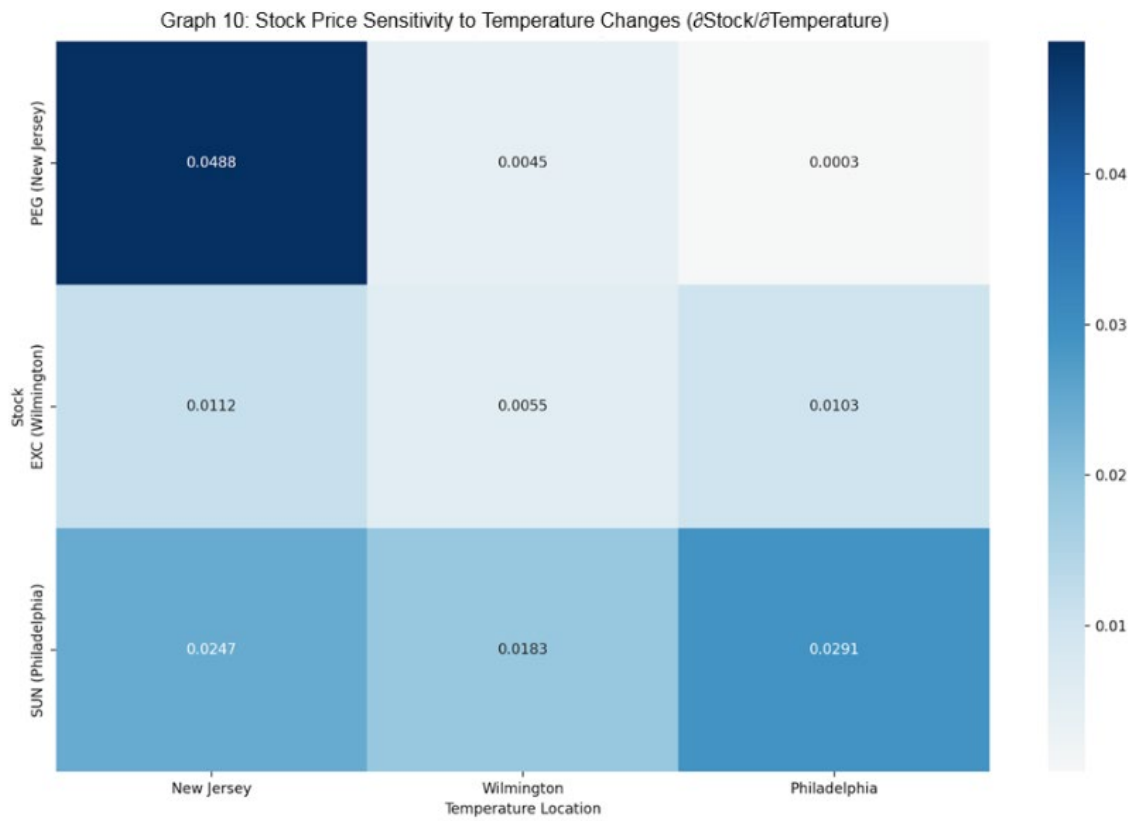
Homoscedasticity (assumption of constant variance) is somewhat violated since the residuals versus time plot shows some pattern, such as the large outlier of about 58.71, and the residual variance is not constant, which may indicate some heteroscedasticity. Nonconstant variance may have some consequences on the accuracy of the predictions over time.

Therefore, the ARIMA models meet the basic requirement of uncorrelated errors and thus are suitable for forecasting stock prices. However, non-normal residuals and heteroscedasticity imply that the corresponding intervals of prediction might also not be very reliable, especially when prices are highly volatile. Further improvement in efficiency may be achieved by incorporation of external variables, such as meteorological conditions, as regressors. Considering the objective of investigating the impact of environmental factors, such as temperature, on stock prices, the inclusion of these elements as exogenous

variables in the ARIMA model may enhance predictive precision, particularly for companies operating in industries sensitive to weather conditions, such as energy.

3.4 Partial Derivatives

Below is the heatmap showing the sensitivity of each stock to temperature changes in New Jersey, Wilmington, and Philadelphia. The values shown are the partial derivatives (∂Stock/∂Temperature) for each stock-temperature pair to show how stock prices vary with temperature changes.



Since the data points are discrete, I approximated this using finite differences:

$$\frac{\partial S}{\partial T} \approx \frac{\Delta S}{\Delta T} = \frac{S_{t+1} - S_t}{T_{t+1} - T_t}$$

Looking at the first two data points from 2020, January 2nd to January 3rd, I can show a sample calculation:

$$\frac{\Delta S}{\Delta T} = \frac{58.70 - 58.71}{30.5 - 38.5} = \frac{-0.01}{-8.0} = 0.00125$$

The average partial derivative across all points is Average Partial Derivative: -0.1366 , which represents the average sensitivity of the stock price to temperature changes in USD/°F.

PEG (New Jersey) reserves the highest regard for its local temperature, with a strong inverse correlation as evidenced by -0.1366 USD/°F. It is also quite interesting to note that PEG is responsive to Wilmington and Philadelphia regional temperatures at a positive coefficient of 0.0252 USD/°F. This explains why PEG has the highest sensitivity to its local temperature, with a large negative partial derivative. Consequently, stock prices tend to increase when temperatures fall.

EXC (Wilmington) does not have large temperature variance relative to PEG and it has an average sensitivity of -0.0117 USD/°F. It follows therefore, that in different markets, EXC has more stable sensitivities.

With respect to stock prices, it appears that there is considerable variability with regards to temperature but within its own region pays off largely (0.0187 USD/°F) while trading elsewhere yields a loss (-0.0175 USD/°F). In this regard, it means that temperatures do affect stock prices but not uniformly across all stocks. Hence, SUN is more sensitive to the temperatures of Philadelphia, at which point stock prices do increase as temperatures increase.

4. Conclusion

4.1 Results

The analysis reveals that temperature impacts on stock prices vary significantly across PEG, EXC, and SUN, each reflecting unique responses to regional climate patterns. PEG's negligible correlation and quantile regression results indicate minimal influence from temperature, suggesting its stock price is driven by broader market factors rather than daily weather. EXC displays some negative sensitivities, especially in the lower quantiles of the stock price, and points toward moderate vulnerability to temperature variability, its ARIMA fit, however, has limited sensitivities and reflects resistance to short-term weather variability. On the other hand, SUN exhibits a consistent positive correlation and sensitivity for quantiles and ARIMA projections, which suggests that higher temperatures could favor its stock price, consistently with seasonal energy demand. Taken together, these results would indicate that, indeed, temperature could affect energy stocks, but the effect is highly specific to the regional and operational context of each company.

4.2 Evaluation

These findings demonstrate that although temperature changes may affect the performance of a number of energy company's stock, the effect is neither consistent nor uncomplicated. The minor relationship of the PEG demonstrates the strength of firms which are less impacted by seasonal energy consumption. SUN, on the other hand, positively responds in situations where the demand for the energy is caused by warm weather conditions, implying that the firm may enjoy some benefits during summer

periods. This disparity emphasizes the necessity of appreciating the detail of company individual operational and market strategies. Turning to the general conclusions, these results indicate that temperature is relevant, but more importantly it works in conjunction with policies of the locality and the degree of technology diffusion within energy firms.

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5. Appendix

Dates	Stock Closing Prices			Average Temperature		
	PEG (New Jersey)	EXC (Wilmington)	SUN (Philadelphia)	Philadelphia	New Jersey	Wilmington
1/2/2020 16:00	58.71	32.41	30.82	35	38.5	35
1/3/2020 16:00	58.7	32.51	30.98	43.5	30.5	43.5
1/6/2020 16:00	58.13	32.55	31.38	40	34.5	40
1/7/2020 16:00	57.6	32.59	31.43	36.5	36	36.5
1/8/2020 16:00	57.96	32.82	31.51	32	34.5	32
1/9/2020 16:00	58.42	32.92	31.34	25.5	30.5	25.5
1/10/2020 16:00	58.3	33	31.3	36	26.5	36
1/13/2020 16:00	58.25	32.95	31.53	39.5	47.5	39.5
1/14/2020 16:00	58.38	32.68	32.11	44	36.5	44
1/15/2020 16:00	59.09	33.21	32.2	44	37.5	44
1/16/2020 16:00	59.02	33.34	32.13	40	40.5	40
1/17/2020 16:00	59.16	33.81	32.58	34.5	33.5	34.5
1/21/2020 16:00	59.55	33.94	32.35	28.5	23.5	24
1/22/2020 16:00	60.3	34	31.96	32	23.5	28.5
1/23/2020 16:00	61.74	34.21	32.11	41	26.5	32
1/24/2020 16:00	61.66	34.05	31.54	44	31	41
1/27/2020 16:00	60.42	33.72	31.57	37	38.5	37
1/28/2020 16:00	59.67	34.05	31.09	37.5	39.5	37.5
1/29/2020 16:00	60.1	34.11	31.67	35	36.5	35
1/30/2020 16:00	60	34.28	31.43	N/A	29.5	N/A
1/31/2020 16:00	59.2	33.95	31.11	36.5	27	36.5
2/3/2020 16:00	59.33	34.38	31.62	45	37	45
2/4/2020 16:00	59.07	34.31	31.78	53	44	53

2/5/2020 16:00	59.28	34.48	31.84	43.5	47	43.5
2/6/2020 16:00	58.33	34.43	31.35	37	36	38
2/7/2020 16:00	57.7	34.38	31.12	43.5	37.5	43.5
2/10/2020 16:00	57.44	34.46	30.2	40	33.5	40
2/11/2020 16:00	57.82	35.2	30.73	45.5	41	45.5
2/12/2020 16:00	58.21	35.14	31.08	39.5	41.5	39.5
2/13/2020 16:00	58.36	35.36	30.92	40.5	39	40.5
2/14/2020 16:00	58.48	35.5	30.94	28	36	28
2/18/2020 16:00	59.02	35.94	30.62	46.5	40	46.5
2/19/2020 16:00	58.9	35.38	30.28	24.5	39.5	41.5
2/20/2020 16:00	59.18	35.56	30.71	24	36	30.5
2/21/2020 16:00	59.96	35.47	31.04	32	26	25.5
2/24/2020 16:00	59.11	34.83	30.71	38	40.5	43.5
2/25/2020 16:00	57.41	33.98	29.35	37	44	44.5
2/26/2020 16:00	55.57	33.56	29.48	35	45.5	45
2/27/2020 16:00	52.87	32.2	27.33	36.5	43.5	N/A
2/28/2020 16:00	51.31	30.75	27.03	33	33.5	33
3/2/2020 16:00	53.71	32.41	28.41	46	33.5	46
3/3/2020 16:00	53.05	31.52	28.05	48.5	43	48.5
3/4/2020 16:00	55.59	33.4	28.65	48	50	48
3/5/2020 16:00	53.99	32.91	27.97	42	44.5	42
3/6/2020 16:00	54.73	32.22	26.26	41.5	39	41.5
3/9/2020 16:00	50.76	29.9	22	N/A	42.5	N/A

3/10/2020 16:00	52.94	30.76	24.86	51.5	52	51.5
3/11/2020 16:00	48.75	28.4	22.55	56.5	50.5	56.5
3/12/2020 16:00	44.59	25.16	17.28	46.5	45	46.5
3/13/2020 16:00	44.14	25.94	19	58	42.5	58
3/16/2020 16:00	40.28	21.76	15.54	40.5	40.5	40.5
3/17/2020 16:00	43.08	25.68	15.33	46.5	37	46.5
3/18/2020 16:00	38.26	23.57	11.93	44.5	45	44.5
3/19/2020 16:00	39.14	23.12	15	45.5	46	45.5
3/20/2020 16:00	37.82	22.11	15.29	62	44.5	62
3/23/2020 16:00	36.86	21.39	13.22	38.5	36.5	38.5
3/24/2020 16:00	40.78	24.62	13.73	45.5	35	45.5
3/25/2020 16:00	40.89	24.65	14.12	42.5	44.5	42.5
3/26/2020 16:00	44.01	26.1	16	44.5	36.5	44.5
3/27/2020 16:00	44.48	25.32	15.62	55	43	55
3/30/2020 16:00	44.51	26.56	14.8	53	45	53
3/31/2020 16:00	44.91	26.26	15.64	42.5	46	42.5
4/1/2020 16:00	42.79	23.88	14.35	45	42	45
4/2/2020 16:00	45.28	24.74	14.81	45	45	45
4/3/2020 16:00	44.36	23.36	14.72	49	47.5	49
4/6/2020 16:00	46.85	25.91	16.06	54.5	54.5	54.5
4/7/2020 16:00	47.53	25.53	16.06	52.5	54.5	52.5
4/8/2020 16:00	50.86	26.81	16.62	59	55	59
4/9/2020 16:00	52.95	27.93	17.85	51	54	51

4/13/2020 16:00	51.66	26.93	17.88	61	48.5	61
4/14/2020 16:00	53.48	27.87	19.02	48.5	55.5	48.5
4/15/2020 16:00	52.01	26.42	19.07	43	46.5	43
4/16/2020 16:00	52.17	26.47	18.88	42	44	42
4/17/2020 16:00	54.62	27.42	20.21	41.5	37	41.5
4/20/2020 16:00	51.84	26.44	20.17	50	49.5	50
4/21/2020 16:00	50.01	25.45	21.07	49.5	45	49.5
4/22/2020 16:00	51.1	26.54	23.05	42	45	42
4/23/2020 16:00	50.37	25.91	23.07	47	41	47
4/24/2020 16:00	51.37	26.11	22.67	46	43.5	46
4/27/2020 16:00	52.01	26.98	23.6	46	43.5	46
4/28/2020 16:00	51.88	26.99	23.78	50	52	50
4/29/2020 16:00	52.3	27.34	24.79	57	51.5	57
4/30/2020 16:00	50.71	26.45	24.82	60	N/A	60
5/1/2020 16:00	49.41	25.54	24.27	59	53	59
5/4/2020 16:00	49.14	25.29	24.44	59.5	65.5	59.5
5/5/2020 16:00	49.84	25.59	24.99	50.5	56.5	50.5
5/6/2020 16:00	48.05	24.59	23.46	46	53	46
5/7/2020 16:00	47.82	25	22.71	51	48	51
5/8/2020 16:00	48.77	26.7	23.84	45	54	45
5/11/2020 16:00	48.2	26.59	24.67	50.5	49	50.5
5/12/2020 16:00	46.8	26.27	23.91	48	52.5	48
5/13/2020 16:00	44.96	25.45	22.06	51	47.5	51

5/14/2020 16:00	46.58	25.75	22.76	58	49	58
5/15/2020 16:00	45.46	25.17	23.67	70	54	70
5/18/2020 16:00	49.35	26.78	23.85	58.5	60	58.5
5/19/2020 16:00	48.5	26.19	24.48	58	60.5	58
5/20/2020 16:00	48.75	26.37	25.22	52.5	56	52.5
5/21/2020 16:00	48.07	25.99	25.29	57	51	57
5/22/2020 16:00	48.12	26.29	25.42	60	55	60
5/26/2020 16:00	49.4	26.82	25.13	66.5	65	56
5/27/2020 16:00	49.91	27.06	25.55	68	70	57
5/28/2020 16:00	51.03	27.29	25.43	71	70	65
5/29/2020 16:00	51.04	27.33	25.8	74.5	69.5	68
6/1/2020 16:00	52.22	27.71	25.55	57.5	58	57.5
6/2/2020 16:00	52.27	27.77	26.37	63	59	63
6/3/2020 16:00	52.75	28.47	26.67	73	61.5	73
6/4/2020 16:00	51.94	27.81	26.65	76.5	69.5	76.5
6/5/2020 16:00	53.48	28.79	26.32	76.5	73	76.5
6/8/2020 16:00	55.5	29.86	27	68	64.5	68
6/9/2020 16:00	53.67	28.9	26.75	71.5	68	71.5
6/10/2020 16:00	52.2	28.53	26.2	75	74	75
6/11/2020 16:00	50	27.14	24.21	73.5	78	73.5
6/12/2020 16:00	49.71	27.04	24.88	76	73.5	76
6/15/2020 16:00	49.9	27.33	25.71	63	61.5	63
6/16/2020 16:00	50.72	27.53	26.47	65	63	65

6/17/2020 16:00	50.46	27.43	26.02	65	65	65
6/18/2020 16:00	50.37	27.25	26.33	N/A	67	N/A
6/19/2020 16:00	48.71	26.77	26.33	73.5	71.5	73.5
6/22/2020 16:00	49.13	26.8	26.2	N/A	74.5	N/A
6/23/2020 16:00	49.04	26.57	26.49	N/A	78.5	N/A
6/24/2020 16:00	48.16	26.02	25.46	N/A	80.5	N/A
6/25/2020 16:00	47.97	25.27	24.77	73.5	75	73.5
6/26/2020 16:00	47.5	24.76	24.33	73.5	72.5	73.5
6/29/2020 16:00	49.21	25.67	23.52	77.5	77	77.5
6/30/2020 16:00	49.16	25.89	22.83	75	73.5	75
7/1/2020 16:00	49.75	26.65	23.26	N/A	71	N/A
7/2/2020 16:00	50.11	26.74	23.27	N/A	72.5	N/A
7/6/2020 16:00	50.62	26.68	23.47	81	79.5	81
7/7/2020 16:00	50.29	26.39	23.4	75.5	81.5	75.5
7/8/2020 16:00	50.87	26.5	23.69	80.5	77	80.5
7/9/2020 16:00	49.79	26.12	23.48	81	77	81
7/10/2020 16:00	50.59	26.67	23.76	77.5	79.5	77.5
7/13/2020 16:00	51.81	26.73	23.34	79	78	79
7/14/2020 16:00	51.52	27.01	23.14	76.5	76	76.5
7/15/2020 16:00	52.4	27.14	23.71	78	74	78
7/16/2020 16:00	52.91	27.15	23.58	77.5	74.5	77
7/17/2020 16:00	55.08	28.1	23.23	N/A	73	77.5
7/20/2020 16:00	52.82	27.34	23.59	75.5	85	85.5

7/21/2020 16:00	52.27	27.63	24.76	75.5	82	85
7/22/2020 16:00	52.99	27.81	24.72	81.5	81.5	81.5
7/23/2020 16:00	52.71	27.56	24.78	79.5	81.5	79.5
7/24/2020 16:00	52.27	26.96	24.54	77.5	80.5	77.5
7/27/2020 16:00	51.99	26.81	24.59	82.5	80	82.5
7/28/2020 16:00	52.41	27.09	24.54	83	81	83
7/29/2020 16:00	52.59	27.51	25.49	81	79	81
7/30/2020 16:00	52.39	27.44	25.35	83	79	83
7/31/2020 16:00	55.94	27.54	24.82	75	80	75
8/3/2020 16:00	54.47	27.1	25.88	81	82.5	81
8/4/2020 16:00	55.65	27.64	26.27	73.5	79	73.5
8/5/2020 16:00	54.58	26.91	27.45	77	72	77
8/6/2020 16:00	54.63	27.04	26.26	73.5	74.5	73.5
8/7/2020 16:00	55.65	27.77	25.94	74	72	74
8/10/2020 16:00	55.35	27.33	25.94	81.5	78	81.5
8/11/2020 16:00	54.47	27.04	25.61	81.5	81	81.5
8/12/2020 16:00	55.53	27.61	26	75	79.5	81.5
8/13/2020 16:00	55.46	27.06	25.68	78	82	75
8/14/2020 16:00	55.17	27.06	26.07	74	74.5	78
8/17/2020 16:00	54.72	26.64	27.01	72	66.5	72
8/18/2020 16:00	54.53	26.44	27.3	73	72	73
8/19/2020 16:00	54.34	26.41	27.22	74	71.5	74
8/20/2020 16:00	53.85	25.9	27.22	71.5	65.5	71.5

8/21/2020 16:00	53.28	26.05	26.94	74	67.5	74
8/24/2020 16:00	53.9	26.89	27.09	78	77	78
8/25/2020 16:00	53.6	26.49	26.99	79.5	78.5	79.5
8/26/2020 16:00	52.45	26.59	26.32	73	76	73
8/27/2020 16:00	52.29	26.32	26.32	80	70	80
8/28/2020 16:00	52.16	26.42	26.47	79	75	79
8/31/2020 16:00	52.24	26.33	26.53	69	66.5	69
9/1/2020 16:00	51	25.92	25.38	70.5	65.5	70.5
9/2/2020 16:00	52.92	27.13	26.18	77	70.5	77
9/3/2020 16:00	52.03	26.6	25.91	78	74	78
9/4/2020 16:00	52.52	26.46	25.67	77	71.5	77
9/8/2020 16:00	52	25.98	25.06	73.5	69.5	73.5
9/9/2020 16:00	53.43	26.16	25.05	74	72.5	74
9/10/2020 16:00	52.3	25.57	24.78	76.5	72.5	76.5
9/11/2020 16:00	52.39	25.56	25.3	73.5	77	73.5
9/14/2020 16:00	52.48	25.87	25.72	66.5	66.5	66.5
9/15/2020 16:00	51.65	25.8	25.42	57.5	60.5	57.5
9/16/2020 16:00	53	25.87	25.37	60.5	55.5	60.5
9/17/2020 16:00	52.24	25.63	25.14	68	57	68
9/18/2020 16:00	51.33	25.02	25.1	62	63.5	62
9/21/2020 16:00	51.17	24.79	25.09	54.5	50	54.5
9/22/2020 16:00	52.47	25.12	24.95	58	51.5	58
9/23/2020 16:00	51.92	24.6	24	63	57.5	63

9/24/2020 16:00	53.47	24.83	24.11	64	64.5	64
9/25/2020 16:00	54.13	25.34	24.32	65.5	63.5	65.5
9/28/2020 16:00	54.35	25.5	24.7	73.5	71.5	73.5
9/29/2020 16:00	54.8	25.2	24.86	71.5	70	71.5
9/30/2020 16:00	54.91	25.51	24.39	61	65.5	61
10/1/2020 16:00	55.07	25.56	24.44	62.5	59	62.5
10/2/2020 16:00	55.95	25.64	24.51	58	60	58
10/5/2020 16:00	56.32	26.24	25.07	58	52.5	58
10/6/2020 16:00	55.9	26.52	24.67	58.5	56	58.5
10/7/2020 16:00	57.03	27	24.73	65	56	65
10/8/2020 16:00	58.4	28.05	25.36	58	62.5	58
10/9/2020 16:00	57.56	27.96	25.25	59.5	52.5	59.5
10/12/2020 16:00	57.76	28.52	25.21	56	55	56
10/13/2020 16:00	57.92	29.23	25.19	58	49	58
10/14/2020 16:00	58.02	29.27	24.85	56.5	53	56.5
10/15/2020 16:00	58.45	29.53	25.04	62	55.5	62
10/16/2020 16:00	59.5	30.03	24.82	55	58	55
10/19/2020 16:00	59.5	29.86	24.68	59.5	48	59.5
10/20/2020 16:00	59.89	30.25	24.92	62	54.5	62
10/21/2020 16:00	59.52	30.06	24.84	64	62.5	64
10/22/2020 16:00	60.87	30.12	25.15	66.5	63	66.5
10/23/2020 16:00	60.92	30.1	25.23	64	64	64
10/26/2020 16:00	60.98	29.75	25.11	53.5	44	53.5

10/27/2020 16:00	61	29.35	25.43	57.5	51.5	57.5
10/28/2020 16:00	58.68	28.23	24.67	61	52.5	61
10/29/2020 16:00	59.72	28.65	25.12	52	52	52
10/30/2020 16:00	58.15	28.46	25.17	44	44.5	44
11/2/2020 16:00	58.49	29.22	25.46	40.5	45	40.5
11/3/2020 16:00	59.32	30.3	25.62	47	39.5	47
11/4/2020 16:00	58.4	30.12	26.38	53	42	53
11/5/2020 16:00	58.79	30.72	25.99	55.5	46.5	55.5
11/6/2020 16:00	58.42	30.56	24.91	58.5	51.5	58.5
11/9/2020 16:00	60.38	30.9	25.21	61.5	57.5	61.5
11/10/2020 16:00	60.73	30.67	26.11	62	58	62
11/11/2020 16:00	60.88	31.27	27.05	63.5	59	63.5
11/12/2020 16:00	59.18	30.46	26.54	55	63	55
11/13/2020 16:00	59.38	30.6	26.98	49	51.5	49
11/16/2020 16:00	59.52	30.92	28.56	48	46.5	48
11/17/2020 16:00	59.05	30.72	29.15	43	40.5	43
11/18/2020 16:00	57.82	30.46	27.95	32.5	38	32.5
11/19/2020 16:00	57.62	30.21	28.95	37.5	28	37.5
11/20/2020 16:00	58.21	29.87	28.81	43	34	43
11/23/2020 16:00	58.42	29.65	28.75	47	49	47
11/24/2020 16:00	59.68	30.5	29.15	40.5	45.5	40.5
11/25/2020 16:00	59.97	30.12	29.15	47.5	39.5	47.5
11/27/2020 13:00	59.35	29.78	28.81	53.5	52.5	53.5

11/30/2020 16:00	58.28	29.3	27.79	55	41	55
12/1/2020 16:00	57.51	29.43	28.22	42.5	54	42.5
12/2/2020 16:00	57.7	29.65	28.16	38	41.5	38
12/3/2020 16:00	57.63	29.28	28.54	41.5	35	41.5
12/4/2020 16:00	57.63	29.2	28.91	44	38.5	44
12/7/2020 16:00	57.53	29.39	28.89	33	32.5	33
12/8/2020 16:00	57.23	29.48	29.29	34	31.5	34
12/9/2020 16:00	56.89	29.31	29.29	34	30.5	34
12/10/2020 16:00	56.6	29.39	30.26	41	33	41
12/11/2020 16:00	56.48	29.38	30.62	45	35.5	45
12/14/2020 16:00	56.43	29.29	29.75	38	47	38
12/15/2020 16:00	57.81	30.47	29.78	32.5	34.5	32.5
12/16/2020 16:00	57.15	30.27	29.05	31	29.5	31
12/17/2020 16:00	57.19	30.42	29.35	27.5	24	27.5
12/18/2020 16:00	56.82	30.37	28.99	27	25	27
12/21/2020 16:00	56.47	29.61	28.86	35.5	30	35.5
12/22/2020 16:00	56.39	29.79	28.47	40	38	40
12/23/2020 16:00	56.53	29.75	28.72	36	37.5	36
12/24/2020 13:00	57.15	29.84	28.58	50.5	36	50.5
12/28/2020 16:00	57.32	30.11	28.26	38.5	30	38.5
12/29/2020 16:00	57.08	29.84	28.39	32.5	39.5	32.5
12/30/2020 16:00	57.36	29.91	28.7	35.5	30	35.5
12/31/2020 16:00	58.3	30.12	28.78	40	34	40

1/4/2021 16:00	55.73	29.37	28.61	38	32.5	38
1/5/2021 16:00	55.74	29.33	29.2	38	38	38
1/6/2021 16:00	56.91	30.28	28.96	37.5	34	37.5
1/7/2021 16:00	56.1	29.98	28.85	36	34.5	36
1/8/2021 16:00	57.31	30.11	28.76	30	31	30
1/11/2021 16:00	56.98	29.86	29.43	32	30.5	32
1/12/2021 16:00	57.35	30.02	29.88	32.5	28	32.5
1/13/2021 16:00	58.68	30.53	29.89	35	32.5	35
1/14/2021 16:00	57.78	30.37	30.21	41	34.5	41
1/15/2021 16:00	59.38	30.93	30.24	44.5	36.5	44.5
1/19/2021 16:00	58.18	30.55	30.51	36.5	36	36.5
1/20/2021 16:00	59.26	30.71	30.79	33.5	32.5	33.5
1/21/2021 16:00	58.35	30.69	31	32	27.5	32
1/22/2021 16:00	58.36	30.35	30.97	39	28.5	39
1/25/2021 16:00	58.52	30.92	30.97	32	24.5	32
1/26/2021 16:00	58.33	30.57	30.75	35	28	35
1/27/2021 16:00	56.36	29.26	30.31	35	30	35
1/28/2021 16:00	57.05	29.99	30.69	26	29.5	26
1/29/2021 16:00	56.43	29.65	31.28	23	21.5	23
2/1/2021 16:00	57.15	30.02	31.87	27.5	N/A	27.5
2/2/2021 16:00	58.35	30.57	31.98	31.5	23.5	31.5
2/3/2021 16:00	58.1	30.3	32.61	31.5	25	31.5
2/4/2021 16:00	58.56	31.15	33	33.5	31.5	33.5

2/5/2021 16:00	59.32	31.16	32.33	39	34.5	39
2/8/2021 16:00	58.81	31.2	32.73	22.5	22	22.5
2/9/2021 16:00	58.73	31.08	32.47	29.5	18.5	29.5
2/10/2021 16:00	58.85	30.9	32.36	30	20	30
2/11/2021 16:00	58.57	30.49	31.8	28.5	24	28.5
2/12/2021 16:00	58.27	30.11	31.59	24.5	23	24.5
2/16/2021 16:00	57.95	29.95	31.76	37.5	32.5	37.5
2/17/2021 16:00	58.01	30.08	31.78	25.5	34.5	25.5
2/18/2021 16:00	58.5	30.46	30.58	25	24.5	25
2/19/2021 16:00	57.75	29.52	31	28	24	28
2/22/2021 16:00	57.09	28.96	31.01	31	24	31
2/23/2021 16:00	57.77	29.1	31.35	40	29.5	40
2/24/2021 16:00	56.05	28.67	31.92	43	34.5	43
2/25/2021 16:00	55.9	27.76	31.31	40	39	40
2/26/2021 16:00	53.83	27.54	30.6	34.5	36.5	34.5
3/1/2021 16:00	55.09	28.04	31.41	40	38	40
3/2/2021 16:00	54.61	28.04	31.56	29	32.5	29
3/3/2021 16:00	54.25	28.2	31.44	40.5	25	40.5
3/4/2021 16:00	54.66	28.37	31.56	37	35.5	37
3/5/2021 16:00	56.26	28.28	31.39	29	30.5	29
3/8/2021 16:00	57.1	29.03	31.59	34	31	34
3/9/2021 16:00	57.53	29.63	31.35	49	32.5	49
3/10/2021 16:00	57.74	29.86	31.81	52	44	52

3/11/2021 16:00	56.78	29.98	32.06	56.5	42	56.5
3/12/2021 16:00	57.69	30.38	32.33	60	55.5	60
3/15/2021 16:00	58.39	30.93	32.51	46	37	33
3/16/2021 16:00	58.56	30.93	32.15	33	29	35.5
3/17/2021 16:00	58.49	30.65	32.15	35.5	30.5	41.5
3/18/2021 16:00	58.57	30.46	31.28	41.5	38.5	45
3/19/2021 16:00	58.91	30.65	31.37	45	39	40
3/22/2021 16:00	57.88	30.52	31.09	40	45.5	49
3/23/2021 16:00	58.32	30.53	31.09	49	45.5	53
3/24/2021 16:00	58.71	30.62	31.49	53	48.5	53
3/25/2021 16:00	59.42	30.94	31.78	53	49.5	55.5
3/26/2021 16:00	59.3	31.05	32.37	55.5	57.5	66
3/29/2021 16:00	60.21	31.21	32.59	66	50.5	47
3/30/2021 16:00	59.81	31.03	32.2	50.5	41.5	50.5
3/31/2021 16:00	60.21	31.2	31.84	N/A	46.5	55
4/1/2021 16:00	59.67	31.32	32.44	44.5	50.5	44.5
4/5/2021 16:00	60.66	31.73	32.52	55	47	44
4/6/2021 16:00	60.92	31.79	32.74	57.5	52	47
4/7/2021 16:00	61.23	32.07	33.03	58.5	53.5	50
4/8/2021 16:00	61.3	32.03	33.09	56	56	46
4/9/2021 16:00	61.25	31.98	32.94	52	55	47
4/12/2021 16:00	61.29	31.92	32.97	57	51	57
4/13/2021 16:00	62.41	32.22	32.88	53	43	53

4/14/2021 16:00	62.82	32.42	32.93	55	52.5	55
4/15/2021 16:00	62.4	32.65	33.61	55	55	55
4/16/2021 16:00	63.35	32.93	33.82	50.5	47.5	50.5
4/19/2021 16:00	63.48	32.46	33.61	53.5	52	53.5
4/20/2021 16:00	63.65	32.67	33.72	61.5	51.5	61.5
4/21/2021 16:00	63.55	32.59	33.73	54.5	56	54.5
4/22/2021 16:00	62.92	32.16	33.65	40.5	52.5	40.5
4/23/2021 16:00	62.94	32.04	33.58	47.5	41.5	47.5
4/26/2021 16:00	62.16	31.85	33.74	50.5	50.5	50.5
4/27/2021 16:00	62.1	31.77	33.92	60.5	49	60.5
4/28/2021 16:00	62.4	31.82	34.48	72	54	72
4/29/2021 16:00	63.09	32.02	35.01	73	65	73
4/30/2021 16:00	63.16	32.06	35.1	58	66.5	58
5/3/2021 16:00	63.65	32.02	35.66	56.5	66.5	56.5
5/4/2021 16:00	63.19	31.94	36.16	70	59.5	70
5/5/2021 16:00	61.31	31.05	36.64	65	64	65
5/6/2021 16:00	62.57	31.73	35.93	55	55.5	55
5/7/2021 16:00	63.15	31.87	36.09	56.5	53	56.5
5/10/2021 16:00	63.28	31.76	35.51	55	53.5	55
5/11/2021 16:00	62.25	31.7	34.83	49.5	52	56.5
5/12/2021 16:00	60.7	30.75	34.64	55	50	54.5
5/13/2021 16:00	62.15	31.68	34.87	56.5	55	59
5/14/2021 16:00	62.52	32.17	35.13	54.5	55.5	61.5

5/17/2021 16:00	62.29	32.17	35.36	59	59	62.5
5/18/2021 16:00	62.11	32.56	35.32	61.5	61.5	66
5/19/2021 16:00	61.98	32.49	35.44	62	65	72
5/20/2021 16:00	62.58	32.63	35.68	60.5	70	67
5/21/2021 16:00	63.02	32.7	35.92	62.5	63.5	68
5/24/2021 16:00	63.46	32.84	36.18	66	71.5	68
5/25/2021 16:00	62.19	32.04	35.5	72	59.5	64.5
5/26/2021 16:00	62.3	32.22	35.25	67	66.5	72.5
5/27/2021 16:00	62.03	31.82	34.92	68	75.5	68.5
5/28/2021 16:00	62.12	32.19	35.04	71.5	67	59.5
6/1/2021 16:00	62.03	32.2	35.58	64.5	59	64.5
6/2/2021 16:00	61.67	32.24	36.24	68	62.5	68
6/3/2021 16:00	61.71	32.65	36.74	70	66	70
6/4/2021 16:00	61.27	32.53	36.45	72.5	65	72.5
6/7/2021 16:00	60.87	32.71	36.85	81	79.5	81
6/8/2021 16:00	61.01	32.64	36.95	77.5	81	77.5
6/9/2021 16:00	61.13	33.07	37.28	78.5	78.5	78.5
6/10/2021 16:00	61.63	33.32	37.87	75.5	78.5	75.5
6/11/2021 16:00	61.8	33.46	38.32	64.5	70.5	64.5
6/14/2021 16:00	61.84	33.61	38.17	73.5	64.5	73.5
6/15/2021 16:00	62.35	33.61	37.95	70.5	69.5	70.5
6/16/2021 16:00	61.23	32.96	38.07	68	67.5	68
6/17/2021 16:00	61.2	32.81	38.08	66.5	61.5	66.5

6/18/2021 16:00	59.34	31.78	36.88	69.5	63.5	69.5
6/21/2021 16:00	60.06	31.89	37.92	76.5	74.5	76.5
6/22/2021 16:00	59.62	31.62	38.14	78.5	78	78.5
6/23/2021 16:00	59.19	31.49	38.12	63	62	63
6/24/2021 16:00	59.37	31.56	37.81	66	62	66
6/25/2021 16:00	59.86	31.84	37.49	67	63.5	67
6/28/2021 16:00	60.24	32.01	36.85	81.5	80.5	81.5
6/29/2021 16:00	59.49	31.79	37.1	82.5	83	82.5
6/30/2021 16:00	59.74	31.61	37.7	84.5	85	84.5
7/1/2021 16:00	60.24	32.09	37.57	77	84	77
7/2/2021 16:00	60.1	32.19	38.08	71.5	76	71.5
7/6/2021 16:00	60.3	32.02	37.9	80.5	74	80.5
7/7/2021 16:00	60.85	32.19	37.78	83	80	83
7/8/2021 16:00	60.54	32.05	37.58	79	80	79
7/9/2021 16:00	60.49	32.04	37.43	77.5	78.5	77.5
7/12/2021 16:00	60.7	32.3	37.15	82	74	82
7/13/2021 16:00	59.83	31.95	37	80.5	78.5	80.5
7/14/2021 16:00	60.39	32.2	36.95	83	71.5	83
7/15/2021 16:00	61.19	32.62	36.32	81	77	81
7/16/2021 16:00	61.74	32.81	36.35	84	79	84
7/19/2021 16:00	60.19	32.33	35.09	77	76	77
7/20/2021 16:00	60.67	32.86	35.7	78.5	76	78.5
7/21/2021 16:00	60.05	32.7	36.31	77.5	76.5	77.5

7/22/2021 16:00	60.21	32.82	36.64	70.5	70.5	70.5
7/23/2021 16:00	61.22	33.19	37.34	73.5	71	73.5
7/26/2021 16:00	61.55	33.11	38.09	79.5	75.5	79.5
7/27/2021 16:00	62.85	33.7	37.22	81.5	76.5	81.5
7/28/2021 16:00	62.5	33.51	37.85	77	78	77
7/29/2021 16:00	63.07	33.76	38.08	74	71.5	74
7/30/2021 16:00	62.23	33.38	38.01	78	70.5	78
8/2/2021 16:00	62.86	33.55	38.07	69	67	69
8/3/2021 16:00	63.51	33.8	38.8	69	67	69
8/4/2021 16:00	63.27	33.36	38.67	70.5	66.5	70.5
8/5/2021 16:00	63.9	34.01	37.46	73.5	68	73.5
8/6/2021 16:00	63.86	33.96	37.02	78	71	78
8/9/2021 16:00	63.83	33.83	36.11	77	73	77
8/10/2021 16:00	62.9	33.93	36.5	80	76	80
8/11/2021 16:00	63.53	34.38	37.04	82.5	77.5	82.5
8/12/2021 16:00	63.85	34.26	37.03	84.5	80	84.5
8/13/2021 16:00	64.72	34.49	36.91	82	81.5	82
8/16/2021 16:00	64.82	34.65	36.57	75.5	71	75.5
8/17/2021 16:00	64.73	34.5	36.59	75.5	71.5	75.5
8/18/2021 16:00	64	34.4	36.42	80.5	75	80.5
8/19/2021 16:00	64.26	34.8	36.16	81.5	78.5	81.5
8/20/2021 16:00	64.95	35.21	36.45	77	78	77
8/23/2021 16:00	64.34	35	37.22	79	71.5	79

8/24/2021 16:00	63.81	34.89	36.79	81	76.5	81
8/25/2021 16:00	64.11	34.88	37.2	80.5	77.5	80.5
8/26/2021 16:00	63.79	34.8	37.1	83	81	83
8/27/2021 16:00	64.17	35.02	37.23	83.5	81.5	83.5
8/30/2021 16:00	64.24	35.13	37.27	80.5	70.5	80.5
8/31/2021 16:00	63.94	34.97	36.8	78.5	78	78.5
9/1/2021 16:00	64.72	35.38	36.91	77.5	75.5	70.5
9/2/2021 16:00	64.68	35.74	37.26	71.5	64.5	68
9/3/2021 16:00	64.14	35.54	37.59	69.5	64	65.5
9/7/2021 16:00	62.98	35.32	37.12	77.5	70.5	72.5
9/8/2021 16:00	64.1	35.92	36.58	75.5	69	75
9/9/2021 16:00	63.32	35.83	37.1	79	71	68.5
9/10/2021 16:00	62.14	35.6	36.94	68	66.5	66.5
9/13/2021 16:00	62.32	36.07	37.1	75.5	70.5	76.5
9/14/2021 16:00	62.57	35.94	36.93	81	73.5	76.5
9/15/2021 16:00	62.8	36.27	37.19	81.5	72	78.5
9/16/2021 16:00	62.61	35.8	37.3	81	76	73.5
9/17/2021 16:00	61.91	35.57	37.01	75.5	71.5	73
9/20/2021 16:00	62.07	35.46	36.9	73.5	62.5	67.5
9/21/2021 16:00	61.7	34.97	37.01	71.5	64	69
9/22/2021 16:00	61.02	35.23	37.3	76	65.5	75
9/23/2021 16:00	60.69	35.42	37.41	79.5	75	69
9/24/2021 16:00	60.7	35.13	37.25	69	67.5	62

9/27/2021 16:00	61.21	34.88	37.55	67.5	61.5	67.5
9/28/2021 16:00	60.36	34.38	37.26	72	67	68
9/29/2021 16:00	61.26	34.69	37.26	65	59.5	59.5
9/30/2021 16:00	60.9	34.48	37.32	63.5	59.5	62
10/1/2021 16:00	60.46	34.09	37.34	64.5	56	57.5
10/4/2021 16:00	61.15	34.49	38	73.5	64.5	70
10/5/2021 16:00	60.85	34.48	38.01	77	63	71
10/6/2021 16:00	60.95	35.03	38.01	66.5	61	71
10/7/2021 16:00	60.46	34.75	38.09	67.5	62	67.5
10/8/2021 16:00	59.89	34.49	38.7	72.5	64.5	67.5
10/11/2021 16:00	59.28	33.93	39.5	65.5	59.5	65
10/12/2021 16:00	59.42	34.45	39.68	67	63	65
10/13/2021 16:00	60.17	35.01	40.13	66.5	62.5	66.5
10/14/2021 16:00	61.41	35.76	41.86	69	63.5	68
10/15/2021 16:00	62.11	35.53	41.6	71.5	67	71
10/18/2021 16:00	61.73	35.59	41.84	57	51	53.5
10/19/2021 16:00	62.45	35.9	42.07	55.5	49.5	56
10/20/2021 16:00	63.4	36.53	42	63.5	53	63
10/21/2021 16:00	63.09	36.43	41.58	68	63	63
10/22/2021 16:00	63.84	36.6	41.74	68	62	60
10/25/2021 16:00	63.93	36.8	41.6	55	48	67.5
10/26/2021 16:00	64.61	37.11	41.35	68.5	63.5	56
10/27/2021 16:00	63.98	37.08	41.04	64	56.5	57

10/28/2021 16:00	64.33	37.63	40.8	57.5	53	55.5
10/29/2021 16:00	63.8	37.94	40.49	58	51	55
11/1/2021 16:00	64.36	38.44	41.45	58.5	56.5	52
11/2/2021 16:00	63.98	38.34	41.6	56	50	45
11/3/2021 16:00	63.37	38.56	41.45	45.5	43	43
11/4/2021 16:00	63.68	38.6	40.09	45.5	39.5	43
11/5/2021 16:00	64.19	38.62	39.7	45	39.5	42.5
11/8/2021 16:00	62.17	37.89	40.12	46	42	54.5
11/9/2021 16:00	62.16	38.59	39.81	57	49.5	59.5
11/10/2021 16:00	62.72	38.84	39.41	61	51.5	57.5
11/11/2021 16:00	62.19	39.02	39.94	58	47	55
11/12/2021 16:00	62.17	38.81	40.3	62	48	53.5
11/15/2021 16:00	63.34	39.07	39.98	43	40	41
11/16/2021 16:00	62.72	38.7	39.96	35	40.5	38.5
11/17/2021 16:00	62.46	38.22	39.87	43	37.5	48
11/18/2021 16:00	62.4	38.19	40.13	57	40.5	55.5
11/19/2021 16:00	63.07	38.24	39.96	56.5	53	39.5
11/22/2021 16:00	63.44	38.49	39.85	47.5	45	41
11/23/2021 16:00	64.2	38.79	40.71	44.5	40	35.5
11/24/2021 16:00	64.59	38.88	41.04	38	34	35.5
11/26/2021 13:00	63.64	38.11	40.37	47	43.5	40.5
11/29/2021 16:00	64.25	38.64	39.51	43.5	33	42
11/30/2021 16:00	62.49	37.62	38.39	40	32.5	39
12/1/2021 16:00	62.44	37.3	37.87	43.5	33	39.5

12/2/2021 16:00	63.17	37.75	38.31	49	36.5	46
12/3/2021 16:00	63.45	37.62	38.14	51.5	46.5	43
12/6/2021 16:00	63.71	38.19	38.12	45.5	39	50.5
12/7/2021 16:00	64.62	38.46	38.21	55	46.5	35
12/8/2021 16:00	63.91	38.11	38.02	39.5	35.5	35.5
12/9/2021 16:00	63.74	37.91	37.55	37	34.5	33
12/10/2021 16:00	63.8	38.46	38.05	41.5	33.5	39
12/13/2021 16:00	65.19	38.62	37.78	50.5	39	42.5
12/14/2021 16:00	64.53	38.41	37.47	46.5	42.5	43.5
12/15/2021 16:00	65.55	39.01	37.57	48.5	40.5	46
12/16/2021 16:00	65.61	38.78	38.06	48.5	41	52
12/17/2021 16:00	64.09	38.18	38.16	57	53.5	52.5
12/20/2021 16:00	64.09	38.09	37.7	39	29	29
12/21/2021 16:00	64.31	38.48	38.54	33	28.5	34
12/22/2021 16:00	64.6	38.79	38.82	39.5	30.5	36.5
12/23/2021 16:00	64.58	38.87	39.11	38	36.5	31
12/27/2021 16:00	65.06	39.1	39.83	49.5	39	35
12/28/2021 16:00	65.58	39.67	39.97	39.5	34.5	41.5
12/29/2021 16:00	65.97	40.41	40.3	45	41.5	45.5
12/30/2021 16:00	66.32	40.91	40.48	47	40.5	47
12/31/2021 16:00	66.73	41.2	40.83	51	44	48